Keywords: data attribution • video generation • optical flow • temporal dynamics

Motion Attribution for Video Generation

MOTIVE Identifies Training Clips That Teach Video Models How *Things Move*

By Xindi Wu, Despoina Paschalidou, Jun Gao, Antonio Torralba, Laura Leal-Taixe, Olga Russakovsky, Sanja Fidler & Jonathan Lorraine (NVIDIA, MIT, Princeton)

arXiv:2601.08828 • ICML 2026 Oral

Data attribution for video generation has focused on visual appearance while ignoring the dimension users care about most: motion. MOTIVE (MOTIon attribution for Video gEneration) introduces motion-weighted influence functions that isolate temporal dynamics from static appearance, identifying which training clips teach models how objects and cameras move. Fine-tuning on MOTIVE-selected data achieves a **74.1%** human preference win rate over the base model and raises VBench dynamic degree from 41.3% to **47.6%**.

AT A GLANCE

Problem: Existing video data attribution methods measure influence on per-frame appearance, not temporal dynamics; data curation cannot target motion quality specifically.

Why it matters: Motion realism is the primary dimension separating high-quality from poor video generation; without motion-specific attribution, curation is blind to it.

Method: Motion-weighted optical-flow loss masks focus gradient-based influence computation on dynamic spatial-temporal regions.

Result: 74.1% human win rate; VBench dynamic degree 47.6% vs. 41.3% (random) and 43.8% (whole-video attribution).

Evidence: Evaluated on CogVideoX fine-tuning with three curated datasets; human studies (n=200); VBench motion metrics.

The Big Picture

Video generation has advanced dramatically, but temporal realism—the physical plausibility of motion—consistently lags behind per-frame visual quality. The gap is a data problem: it is unclear which training clips are responsible for teaching a model that objects fall, water flows, or cameras pan smoothly.

Data attribution, which quantifies each training example's influence on model behavior, is the natural tool for this problem. But existing attribution methods compute influence on a generic generation loss, dominated by the high-frequency spatial structure of individual frames. Motion signals—which are lower-magnitude and distributed across frame pairs—are drowned out.

MOTIVE addresses this by constructing a motion-specific loss function and computing influence with respect to it.

Why Existing Approaches Fall Short

Gradient-based influence functions (TRAK, DataInf, EK-FAC) assign each training example a score based on its gradient alignment with a test-set loss. For video, the default loss is reconstruction error summed uniformly over all pixels and frames.

This aggregation is problematic for motion attribution because:

- Static backgrounds contribute most pixels and dominate the loss.
- Motion signals are typically localized to a small fraction of the spatial extent (moving objects, hands, cameras).
- Two clips with identical backgrounds but opposite motion patterns have near-identical influence scores under whole-video attribution.

Empirically, whole-video attribution selects clips that match the test sequence's scene composition, not its motion patterns, yielding only marginal improvements in dynamic degree (43.8% vs. 41.3% random).

Core Mechanism

MOTIVE constructs a **motion-weighted loss mask** from optical flow: pixels with high inter-frame displacement receive high weight, static pixels receive near-zero weight. Influence computation then operates on this masked loss, making gradients sensitive to motion-generating patterns rather than static appearance.

The motion mask $M(x,t)$ is derived from the optical flow field $f(x,t) \in \mathbb{R}^2$ between frames t and t+1:

$$M(x,t) = \mathrm{softmax}_\tau(\|f(x,t)\|_2)$$

where the softmax is taken over all spatial positions x and times t, with temperature τ controlling mask sharpness. This reweights the loss to be high wherever the scene is moving.

Technical Formulation

The motion-weighted loss for training clip z_i is:

$$\mathcal{L}_M(z_i;\theta) = \sum_{t=1}^{T} \sum_{x \in \Omega} M(x,t) \cdot \ell(v_\theta(x,t),\ f_t(x))$$

where $v_\theta(x,t)$ is the model's prediction for pixel x at frame t and ℓ is the per-pixel reconstruction loss.

The influence score of training clip z_i on a motion-quality test loss $\mathcal{L}_M(z_{\text{test}})$ is:

$$\mathcal{I}(z_i) = -\nabla_\theta \mathcal{L}_M(z_{\text{test}})^\top H_\theta^{-1} \nabla_\theta \mathcal{L}_M(z_i)$$

where $H_\theta = \frac{1}{n}\sum_{j=1}^n \nabla_\theta^2 \mathcal{L}_M(z_j)$ is the Hessian. The Hessian-vector product $H_\theta^{-1}v$ is approximated via 3 iterations of LiSSA (Linear time Stochastic Second-order Algorithm), making MOTIVE scalable to models with billions of parameters.

Learning or Inference Procedure

1. **Flow estimation:** compute optical flow $f(x,t)$ for all frames of each candidate training clip using a pre-trained flow network (RAFT).
2. **Mask construction:** compute $M(x,t)$ via spatiotemporal softmax over flow magnitudes.
3. **LiSSA precomputation:** compute $v_{\text{test}} = H_\theta^{-1}\nabla_\theta \mathcal{L}_M(z_{\text{test}})$ using 3 LiSSA iterations.
4. **Influence scoring:** compute $\mathcal{I}(z_i) = -v_{\text{test}}^\top \nabla_\theta \mathcal{L}_M(z_i)$ for each candidate clip.
5. **Curation:** select top-k clips by influence score for fine-tuning.
6. **Fine-tuning:** standard video diffusion fine-tuning on selected clips.

What the Guarantee Says

MOTIVE provides no formal approximation guarantee on influence estimation, but establishes empirically that the LiSSA approximation error is negligible relative to the rank-ordering of training clips: Spearman rank correlation between 3-iteration and exact influence (computed on small-scale models where exact inversion is tractable) is 0.94.

Experimental Findings

Curation Method	VBench Dyn.	Human Pref.
Random selection	41.3%	–
Whole-video attr.	43.8%	52.1%
MOTIVE	**47.6%**	**74.1%**

Motion smoothness (VBench) also improves: MOTIVE achieves 96.8% vs. 95.2% for random selection. Physical plausibility (human study on gravity and collision realism) improves by 18 percentage points.

Ablations and Interpretation

Flow threshold: Setting $\tau \to 0$ (binary motion mask at median flow magnitude) reduces VBench dynamic degree to 45.1%; smooth soft masks are important.

LiSSA steps: 1 step yields 46.1% (noisy influence estimates); 3 steps yields 47.6%; 10 steps yields 47.7% (negligible gain).

Test set choice: Influence is computed relative to a motion-diverse test set; using a static-scene test set produces data selection equivalent to whole-video attribution, confirming the motion mask is the active ingredient.

Model agnosticism: Results replicate on CogVideoX-2B and CogVideoX-5B; the framework is model-architecture agnostic.

Reference

Xindi Wu, Despoina Paschalidou, Jun Gao, Antonio Torralba, Laura Leal-Taixe, Olga Russakovsky, Sanja Fidler, Jonathan Lorraine. **Motion Attribution for Video Generation**. arXiv:2601.08828, January 2026. https://arxiv.org/abs/2601.08828